

Администрирование локальных сетей

Лабораторная работа № 4. Первоначальное конфигурирование сети

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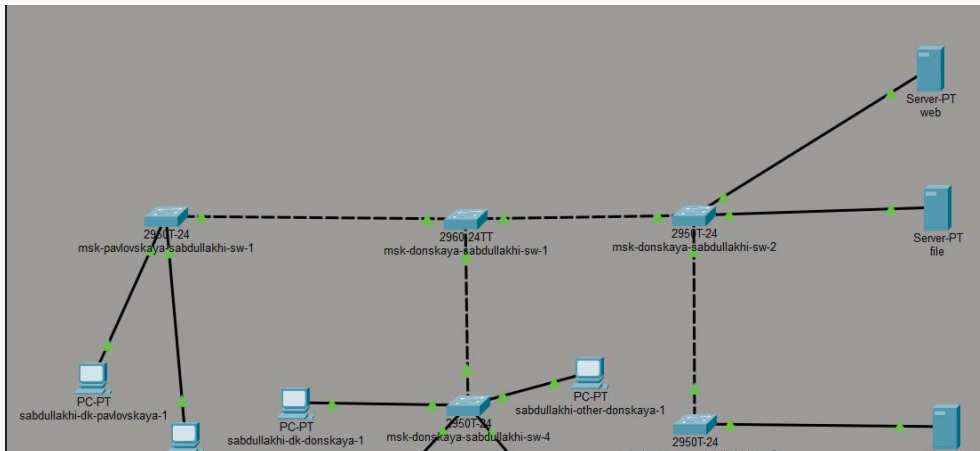
03 марта 2026

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Целью данной лабораторной работы является выполнение первоначальной настройки сетевых коммутаторов в соответствии с заданной топологией L1. Под первоначальной настройкой понимается: присвоение устройству уникального имени; назначение IP адреса для управления устройством; настройка шлюза по умолчанию; конфигурирование парольной защиты на консольном порту и виртуальных терминалах (VTY); настройка защищенного удаленного доступа по протоколу SSH.

Построение топологии сети

Использованные коммутаторы: - msk pavlovskaya sabdullakhi sw 1 - msk donsкаya sabdullakhi sw 1 - msk donsкаya sabdullakhi sw 2 - msk donsкаya sabdullakhi sw 3 - msk donsкаya sabdullakhi sw 4



Настройка msk donskeya sabdullakhi sw 1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskeya-sabdullakhi-sw-1
msk-donskeya-sabdullakhi-sw-1(config)#interface vlan2
msk-donskeya-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to
up

msk-donskeya-sabdullakhi-sw-1(config-if)#no shutdown
msk-donskeya-sabdullakhi-sw-1(config-if)#ip address 10.128.1.2
255.255.255.0
msk-donskeya-sabdullakhi-sw-1(config-if)#exit
msk-donskeya-sabdullakhi-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskeya-sabdullakhi-sw-1(config)#line vty 0 4
msk-donskeya-sabdullakhi-sw-1(config-line)#password cisco
msk-donskeya-sabdullakhi-sw-1(config-line)#login
msk-donskeya-sabdullakhi-sw-1(config-line)#exit
msk-donskeya-sabdullakhi-sw-1(config)#line console 0
msk-donskeya-sabdullakhi-sw-1(config-line)#password cisco
msk-donskeya-sabdullakhi-sw-1(config-line)#login
msk-donskeya-sabdullakhi-sw-1(config-line)#exit
msk-donskeya-sabdullakhi-sw-1(config)#enable secret cisco
msk-donskeya-sabdullakhi-sw-1(config)#service password-encryption
msk-donskeya-sabdullakhi-sw-1(config)#username admin privilege 1 secret
cisco
msk-donskeya-sabdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskeya-sabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskeya-sabdullakhi-
sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskeya-sabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:22:23.303: RSA key size needs to be at least 768 bits for ssh
version 2
*Mar 1 0:22:23.303: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskeya-sabdullakhi-sw-1(config-line)#transport input ssh
msk-donskeya-sabdullakhi-sw-1(config-line)#exit
msk-donskeya-sabdullakhi-sw-1(config)#exit
msk-donskeya-sabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskeya-sabdullakhi-sw-1#write memory
Building configuration...
[OK]
msk-donskeya-sabdullakhi-sw-1#
```

Настройка msk donskeya sabdullakhi sw 2

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-sabdullakhi-sw-2
msk-donskaya-sabdullakhi-sw-2(config)#interface vlan2
msk-donskaya-sabdullakhi-sw-2(config-if)#
%LINK-S-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-sabdullakhi-sw-2(config-if)#no shutdown
msk-donskaya-sabdullakhi-sw-2(config-if)#ip address 10.128.1.3
255.255.255.0
msk-donskaya-sabdullakhi-sw-2(config-if)#exit
msk-donskaya-sabdullakhi-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-sabdullakhi-sw-2(config)#line vty 0 4
msk-donskaya-sabdullakhi-sw-2(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-2(config-line)#login
msk-donskaya-sabdullakhi-sw-2(config-line)#exit
msk-donskaya-sabdullakhi-sw-2(config)#line console 0
msk-donskaya-sabdullakhi-sw-2(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-2(config-line)#login
msk-donskaya-sabdullakhi-sw-2(config-line)#exit
msk-donskaya-sabdullakhi-sw-2(config)#enable secret cisco
msk-donskaya-sabdullakhi-sw-2(config)#service password-encryption
msk-donskaya-sabdullakhi-sw-2(config)#username admin privilege 1 secret
cisco
msk-donskaya-sabdullakhi-sw-2(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-sabdullakhi-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-sabdullakhi-
sw-2.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sabdullakhi-sw-2(config)#line vty 0 4
*Mar 1 0:30:26.21: RSA key size needs to be at least 768 bits for ssh
version 2
*Mar 1 0:30:26.21: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sabdullakhi-sw-2(config-line)#transport input ssh
msk-donskaya-sabdullakhi-sw-2(config-line)#exit
msk-donskaya-sabdullakhi-sw-2(config)#exit
msk-donskaya-sabdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sabdullakhi-sw-2#write memory
Building configuration...
[OK]
```

Настройка msk donsкаya sabdullakhi sw 3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-sabdullakhi-sw-3
msk-donskaya-sabdullakhi-sw-3(config)#interface vlan2
msk-donskaya-sabdullakhi-sw-3(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-sabdullakhi-sw-3(config-if)#no shutdown
msk-donskaya-sabdullakhi-sw-3(config-if)#ip address 10.128.1.4
255.255.255.0
msk-donskaya-sabdullakhi-sw-3(config-if)#exit
msk-donskaya-sabdullakhi-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-sabdullakhi-sw-3(config)#line vty 0 4
msk-donskaya-sabdullakhi-sw-3(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-3(config-line)#login
msk-donskaya-sabdullakhi-sw-3(config-line)#exit
msk-donskaya-sabdullakhi-sw-3(config)#line console 0
msk-donskaya-sabdullakhi-sw-3(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-3(config-line)#login
msk-donskaya-sabdullakhi-sw-3(config-line)#exit
msk-donskaya-sabdullakhi-sw-3(config)#enable secret cisco
msk-donskaya-sabdullakhi-sw-3(config)#service password-encryption
msk-donskaya-sabdullakhi-sw-3(config)#username admin privilege 1 secret
cisco
msk-donskaya-sabdullakhi-sw-3(config)#ip domain-name donsкаya-rudn.edu
msk-donskaya-sabdullakhi-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-sabdullakhi-sw-3.donsкаya-
rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sabdullakhi-sw-3(config)#line vty 0 4
*Mar 1 0:33:6.865: RSA key size needs to be at least 768 bits for ssh
version 2
*Mar 1 0:33:6.865: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sabdullakhi-sw-3(config-line)#transport input ssh
msk-donskaya-sabdullakhi-sw-3(config-line)#exit
msk-donskaya-sabdullakhi-sw-3(config)#exit
msk-donskaya-sabdullakhi-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sabdullakhi-sw-3#write memory
Building configuration...
[OK]
```

Настройка msk donskeya sabdullakhi sw 4

```
Switch>enable
Switch#configure terminal
^
% Invalid input detected at '^' marker.

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-sabdullakhi-sw-4
msk-donskaya-sabdullakhi-sw-4(config)#interface vlan2
msk-donskaya-sabdullakhi-sw-4(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-sabdullakhi-sw-4(config-if)#no shutdown
msk-donskaya-sabdullakhi-sw-4(config-if)#ip address 10.128.1.5
255.255.255.0
msk-donskaya-sabdullakhi-sw-4(config-if)#exit
msk-donskaya-sabdullakhi-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-sabdullakhi-sw-4(config)#line vty 0 4
msk-donskaya-sabdullakhi-sw-4(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-4(config-line)#login
msk-donskaya-sabdullakhi-sw-4(config-line)#exit
msk-donskaya-sabdullakhi-sw-4(config)#line console 0
msk-donskaya-sabdullakhi-sw-4(config-line)#password cisco
msk-donskaya-sabdullakhi-sw-4(config-line)#login
msk-donskaya-sabdullakhi-sw-4(config-line)#exit
msk-donskaya-sabdullakhi-sw-4(config)#enable secret cisco
msk-donskaya-sabdullakhi-sw-4(config)#service password-encryption
msk-donskaya-sabdullakhi-sw-4(config)#username admin privilege 1 secret
cisco
msk-donskaya-sabdullakhi-sw-4(config)#ip domain-name donskeya-rudn.edu
msk-donskaya-sabdullakhi-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-sabdullakhi-sw-4.donskeya-
rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-sabdullakhi-sw-4(config)#line vty 0 4
*Mar 1 0:44:10.280: RSA Key size needs to be at least 768 bits for ssh
version 2
*Mar 1 0:44:10.280: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-sabdullakhi-sw-4(config-line)#transport input ssh
msk-donskaya-sabdullakhi-sw-4(config-line)#exit
msk-donskaya-sabdullakhi-sw-4(config)#exit
msk-donskaya-sabdullakhi-sw-4#
%SYS-5-CONFIG I: Configured from console by console

msk-donskaya-sabdullakhi-sw-4#write memory
Building configuration...
[OK]
```

Настройка msk pavlovskaya sabdullakhi sw 1

```
msk-pavlovskaya-sabdullakhi-sw-1(config)#interface vlan2
msk-pavlovskaya-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to
up

msk-pavlovskaya-sabdullakhi-sw-1(config-if)#no shutdown
msk-pavlovskaya-sabdullakhi-sw-1(config-if)#ip address 10.128.1.6
255.255.255.0
msk-pavlovskaya-sabdullakhi-sw-1(config-if)#exit
msk-pavlovskaya-sabdullakhi-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-sabdullakhi-sw-1(config)#lint vty 0 4
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-sabdullakhi-sw-1(config)#line vty 0 4
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#password cisco
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#login
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-sabdullakhi-sw-1(config)#line console 0
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#password cisco
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#login
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-sabdullakhi-sw-1(config)#enable secret cisco
msk-pavlovskaya-sabdullakhi-sw-1(config)#service password-encryption
msk-pavlovskaya-sabdullakhi-sw-1(config)#username admin privilege 1
secret cisco
msk-pavlovskaya-sabdullakhi-sw-1(config)#ip domain-name
donskaya.rudn.edu
msk-pavlovskaya-sabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-sabdullakhi-
sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-sabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:51:5.235: RSA key size needs to be at least 768 bits for ssh
version 2
*Mar 1 0:51:5.235: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#transport input ssh
msk-pavlovskaya-sabdullakhi-sw-1(config-line)#exit
msk-pavlovskaya-sabdullakhi-sw-1(config)#exit
msk-pavlovskaya-sabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-sabdullakhi-sw-1#write memory
Building configuration...
[OK]
```

В ходе лабораторной работы выполнено первоначальное конфигурирование пяти коммутаторов согласно схеме сети L1. Каждому устройству присвоено уникальное имя, назначен IP-адрес управления из подсети 10.128.1.0/24, настроен шлюз по умолчанию.

Реализована защита доступа: установлены пароли на console и VTY, настроен привилегированный режим, включено шифрование. Для безопасного удалённого управления на всех коммутаторах настроен доступ по SSH.

Все конфигурации сохранены. Коммутаторы готовы к удалённому администрированию.

Спасибо за внимание
